202. The sum of s and 13 is 68.

203. The difference of d and 30 is equal to 52.

204. The difference of c and 25 is equal to 75.

205. 12 less than u is 89.

206. 19 less than w is 56.

207. 325 less than c gives 799.

208. 299 less than d gives 850.

Everyday Math

209. **Insurance** Vince's car insurance has a $\$500$ deductible. Find the amount the insurance company will pay, p, for an $\$1800$ claim by solving the equation $500 + p = 1800$.

210. **Insurance** Marta's homeowner's insurance policy has a $\$750$ deductible. The insurance company paid $\$5800$ to repair damages caused by a storm. Find the total cost of the storm damage, d, by solving the equation $d - 750 = 5800$.

211. **Sale purchase** Arthur bought a suit that was on sale for $\$120$ off. He paid $\$340$ for the suit. Find the original price, p, of the suit by solving the equation $p - 120 = 340$.

212. **Sale purchase** Rita bought a sofa that was on sale for $\$1299$. She paid a total of $\$1409$, including sales tax. Find the amount of the sales tax, t, by solving the equation $1299 + t = 1409$.

Writing Exercises

213. Is $x = 1$ a solution to the equation $8x - 2 = 16 - 6x$? How do you know?

214. Write the equation $y - 5 = 21$ in words. Then make up a word problem for this equation.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
determine whether a number is a solution of an equation.			
model the subtraction property of equality.			
solve equations using the subtraction property of equality.			
solve equations using the addition property of equality.			
translate word phrases to algebraic equations.			
translate to an equation and solve.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

2.4 Find Multiples and Factors

Learning Objectives

By the end of this section, you will be able to:

- Identify multiples of numbers
- Use common divisibility tests
- Find all the factors of a number
- Identify prime and composite numbers

☑ **BE PREPARED** 2.10 Before you get started, take this readiness quiz.

Which of the following numbers are counting numbers (natural numbers)?
$0, 4, 215$
If you missed this problem, review Example 1.1.

☑ **BE PREPARED** 2.11 Find the sum of $3, 5$, and 7.
If you missed the problem, review Example 2.1.

Identify Multiples of Numbers

Annie is counting the shoes in her closet. The shoes are matched in pairs, so she doesn't have to count each one. She counts by twos: $2, 4, 6, 8, 10, 12$. She has 12 shoes in her closet.

The numbers $2, 4, 6, 8, 10, 12$ are called multiples of 2. Multiples of 2 can be written as the product of a counting number and 2. The first six multiples of 2 are given below.

$$1 \cdot 2 = 2$$
$$2 \cdot 2 = 4$$
$$3 \cdot 2 = 6$$
$$4 \cdot 2 = 8$$
$$5 \cdot 2 = 10$$
$$6 \cdot 2 = 12$$

A **multiple of a number** is the product of the number and a counting number. So a multiple of 3 would be the product of a counting number and 3. Below are the first six multiples of 3.

$$1 \cdot 3 = 3$$
$$2 \cdot 3 = 6$$
$$3 \cdot 3 = 9$$
$$4 \cdot 3 = 12$$
$$5 \cdot 3 = 15$$
$$6 \cdot 3 = 18$$

We can find the multiples of any number by continuing this process. Table 2.8 shows the multiples of 2 through 9 for the first twelve counting numbers.

Counting Number	1	2	3	4	5	6	7	8	9	10	11	12
Multiples of 2	2	4	6	8	10	12	14	16	18	20	22	24
Multiples of 3	3	6	9	12	15	18	21	24	27	30	33	36
Multiples of 4	4	8	12	16	20	24	28	32	36	40	44	48
Multiples of 5	5	10	15	20	25	30	35	40	45	50	55	60
Multiples of 6	6	12	18	24	30	36	42	48	54	60	66	72
Multiples of 7	7	14	21	28	35	42	49	56	63	70	77	84
Multiples of 8	8	16	24	32	40	48	56	64	72	80	88	96
Multiples of 9	9	18	27	36	45	54	63	72	81	90	99	108

Table 2.8

Multiple of a Number

A number is a multiple of n if it is the product of a counting number and n.

Recognizing the patterns for multiples of $2, 5, 10,$ and 3 will be helpful to you as you continue in this course.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Multiples" will help you develop a better understanding of multiples.

Figure 2.6 shows the counting numbers from 1 to 50. Multiples of 2 are highlighted. Do you notice a pattern?

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Figure 2.6 Multiples of 2 between 1 and 50

The last digit of each highlighted number in Figure 2.6 is either $0, 2, 4, 6,$ or 8. This is true for the product of 2 and any counting number. So, to tell if any number is a multiple of 2 look at the last digit. If it is $0, 2, 4, 6,$ *or* 8, then the number is a multiple of 2.

EXAMPLE 2.40

Determine whether each of the following is a multiple of 2:

ⓐ 489 ⓑ 3,714

Solution

ⓐ

Is 489 a multiple of 2?	
Is the last digit 0, 2, 4, 6, or 8?	No.
	489 is not a multiple of 2.

ⓑ

Is 3,714 a multiple of 2?	
Is the last digit 0, 2, 4, 6, or 8?	Yes.
	3,714 is a multiple of 2.

TRY IT 2.79 Determine whether each number is a multiple of 2:

ⓐ 678 ⓑ 21,493

TRY IT 2.80 Determine whether each number is a multiple of 2:

ⓐ 979 ⓑ 17,780

Now let's look at multiples of **5**. Figure 2.7 highlights all of the multiples of **5** between **1** and **50**. What do you notice about the multiples of **5**?

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Figure 2.7 Multiples of 5 between 1 and 50

All multiples of 5 end with either 5 or 0. Just like we identify multiples of 2 by looking at the last digit, we can identify multiples of 5 by looking at the last digit.

EXAMPLE 2.41

Determine whether each of the following is a multiple of 5:

ⓐ 579 ⓑ 880

Solution

ⓐ

Is 579 a multiple of 5?	
Is the last digit 5 or 0?	No.
	579 is not a multiple of 5.

ⓑ

Is 880 a multiple of 5?	
Is the last digit 5 or 0?	Yes.
	880 is a multiple of 5.

TRY IT 2.81 Determine whether each number is a multiple of 5.

ⓐ 675 ⓑ 1,578

TRY IT 2.82 Determine whether each number is a multiple of 5.

ⓐ 421 ⓑ 2,690

Figure 2.8 highlights the multiples of 10 between 1 and 50. All multiples of 10 all end with a zero.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Figure 2.8 Multiples of 10 between 1 and 50

EXAMPLE 2.42

Determine whether each of the following is a multiple of 10:

ⓐ 425 ⓑ 350

Solution

ⓐ

Is 425 a multiple of 10?	
Is the last digit zero?	No.
	425 is not a multiple of 10.

ⓑ

Is 350 a multiple of 10?	
Is the last digit zero?	Yes.
	350 is a multiple of 10.

TRY IT 2.83 Determine whether each number is a multiple of 10:

ⓐ 179 ⓑ 3,540

TRY IT 2.84 Determine whether each number is a multiple of 10:

ⓐ 110 ⓑ 7,595

Figure 2.9 highlights multiples of 3. The pattern for multiples of 3 is not as obvious as the patterns for multiples of 2, 5, and 10.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

Figure 2.9 Multiples of 3 between 1 and 50

Unlike the other patterns we've examined so far, this pattern does not involve the last digit. The pattern for multiples of 3 is based on the sum of the digits. If the sum of the digits of a number is a multiple of 3, then the number itself is a multiple of 3. See Table 2.9.

Multiple of 3	3	6	9	12	15	18	21	24
Sum of digits	3	6	9	$1+2$ 3	$1+5$ 6	$1+8$ 9	$2+1$ 3	$2+4$ 6

Table 2.9

Consider the number 42. The digits are 4 and 2, and their sum is $4+2=6$. Since 6 is a multiple of 3, we know that 42 is also a multiple of 3.

EXAMPLE 2.43

Determine whether each of the given numbers is a multiple of 3:

ⓐ 645 ⓑ 10,519

Solution

ⓐ Is 645 a multiple of 3?

Find the sum of the digits.	$6+4+5=15$
Is 15 a multiple of 3?	Yes.
If we're not sure, we could add its digits to find out. We can check it by dividing 645 by 3.	$645 \div 3$
The quotient is 215.	$3 \cdot 215 = 645$

ⓑ Is $10{,}519$ a multiple of 3?

Find the sum of the digits.	$1+0+5+1+9=16$
Is 16 a multiple of 3?	No.
So 10,519 is not a multiple of 3 either..	$645 \div 3$
We can check this by dividing by 10,519 by 3.	$3\overline{)10{,}519}$ $3{,}506\text{R}1$

When we divide $10{,}519$ by 3, we do not get a counting number, so $10{,}519$ is not the product of a counting number and 3. It is not a multiple of 3.

TRY IT 2.85 Determine whether each number is a multiple of 3:

ⓐ 954 ⓑ $3{,}742$

TRY IT 2.86 Determine whether each number is a multiple of 3:

ⓐ 643 ⓑ $8{,}379$

Look back at the charts where you highlighted the multiples of 2, of 5, and of 10. Notice that the multiples of 10 are the numbers that are multiples of both 2 and 5. That is because $10 = 2 \cdot 5$. Likewise, since $6 = 2 \cdot 3$, the multiples of 6 are the numbers that are multiples of both 2 and 3.

Use Common Divisibility Tests

Another way to say that 375 is a multiple of 5 is to say that 375 is divisible by 5. In fact, $375 \div 5$ is 75, so 375 is $5 \cdot 75$. Notice in Example 2.43 that $10{,}519$ is not a multiple 3. When we divided $10{,}519$ by 3 we did not get a counting number, so $10{,}519$ is not divisible by 3.

Divisibility

If a number m is a multiple of n, then we say that m is divisible by n.

Since multiplication and division are inverse operations, the patterns of multiples that we found can be used as divisibility tests. Table 2.10 summarizes divisibility tests for some of the counting numbers between one and ten.

Divisibility Tests	
A number is divisible by	
2	if the last digit is **0**, **2**, **4**, **6**, **or 8**
3	if the sum of the digits is divisible by **3**
5	if the last digit is **5** or **0**
6	if divisible by both **2** and **3**
10	if the last digit is **0**

Table 2.10

EXAMPLE 2.44

Determine whether 1,290 is divisible by 2, 3, 5, and 10.

Solution

Table 2.11 applies the divisibility tests to 1,290. In the far right column, we check the results of the divisibility tests by seeing if the quotient is a whole number.

Divisible by...?	Test	Divisible?	Check
2	Is last digit **0**, **2**, **4**, **6**, **or 8**? *Yes.*	yes	$1290 \div 2 = 645$
3	Is sum of digits divisible by **3**? $1 + 2 + 9 + 0 = 12$ *Yes.*	yes	$1290 \div 3 = 430$
5	Is last digit **5** or **0**? *Yes.*	yes	$1290 \div 5 = 258$
10	Is last digit **0**? *Yes.*	yes	$1290 \div 10 = 129$

Table 2.11

Thus, 1,290 is divisible by 2, 3, 5, and 10.

TRY IT 2.87 Determine whether the given number is divisible by 2, 3, 5, and 10.

6240

TRY IT 2.88 Determine whether the given number is divisible by 2, 3, 5, and 10.

7248

EXAMPLE 2.45

Determine whether 5,625 is divisible by 2, 3, 5, and 10.

Solution

Table 2.12 applies the divisibility tests to 5,625 and tests the results by finding the quotients.

Divisible by...?	Test	Divisible?	Check
2	Is last digit **0**, **2**, **4**, **6**, **or 8**? *No.*	no	$5625 \div 2 = 2812.5$
3	Is sum of digits divisible by **3**? $5+6+2+5=18$ *Yes.*	yes	$5625 \div 3 = 1875$
5	Is last digit is **5** or **0**? *Yes.*	yes	$5625 \div 5 = 1125$
10	Is last digit **0**? *No.*	no	$5625 \div 10 = 562.5$

Table 2.12

Thus, $5{,}625$ is divisible by 3 and 5, but not 2, or 10.

TRY IT 2.89 Determine whether the given number is divisible by $2, 3, 5,$ and 10.

4962

TRY IT 2.90 Determine whether the given number is divisible by $2, 3, 5,$ and 10.

3765

Find All the Factors of a Number

There are often several ways to talk about the same idea. So far, we've seen that if m is a multiple of n, we can say that m is divisible by n. We know that 72 is the product of 8 and 9, so we can say 72 is a multiple of 8 and 72 is a multiple of 9. We can also say 72 is divisible by 8 and by 9. Another way to talk about this is to say that 8 and 9 are factors of 72. When we write $72 = 8 \cdot 9$ we can say that we have factored 72.

$$\underbrace{8 \cdot 9}_{\textit{factors}} = \underbrace{72}_{\textit{product}}$$

Factors

In the expression $a \cdot b$, both a and b are called **factors**. If $a \cdot b = m$, and both a and b are integers, then a and b are factors of m, and m is the product of a and b.

In algebra, it can be useful to determine all of the factors of a number. This is called factoring a number, and it can help us solve many kinds of problems.

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity "Model Multiplication and Factoring" will help you develop a better understanding of multiplication and factoring.

For example, suppose a choreographer is planning a dance for a ballet recital. There are 24 dancers, and for a certain scene, the choreographer wants to arrange the dancers in groups of equal sizes on stage.

In how many ways can the dancers be put into groups of equal size? Answering this question is the same as identifying the factors of 24. Table 2.13 summarizes the different ways that the choreographer can arrange the dancers.

Number of Groups	Dancers per Group	Total Dancers
1	24	$1 \cdot 24 = 24$
2	12	$2 \cdot 12 = 24$
3	8	$3 \cdot 8 = 24$
4	6	$4 \cdot 6 = 24$
6	4	$6 \cdot 4 = 24$
8	3	$8 \cdot 3 = 24$
12	2	$12 \cdot 2 = 24$
24	1	$24 \cdot 1 = 24$

Table 2.13

What patterns do you see in Table 2.13? Did you notice that the number of groups times the number of dancers per group is always 24? This makes sense, since there are always 24 dancers.

You may notice another pattern if you look carefully at the first two columns. These two columns contain the exact same set of numbers—but in reverse order. They are mirrors of one another, and in fact, both columns list all of the factors of 24, which are:

$$1, 2, 3, 4, 6, 8, 12, 24$$

We can find all the factors of any counting number by systematically dividing the number by each counting number, starting with 1. If the quotient is also a counting number, then the divisor and the quotient are factors of the number. We can stop when the quotient becomes smaller than the divisor.

HOW TO

Find all the factors of a counting number.

Step 1. Divide the number by each of the counting numbers, in order, until the quotient is smaller than the divisor.
- If the quotient is a counting number, the divisor and quotient are a pair of factors.
- If the quotient is not a counting number, the divisor is not a factor.

Step 2. List all the factor pairs.
Step 3. Write all the factors in order from smallest to largest.

EXAMPLE 2.46

Find all the factors of 72.

Solution

Divide 72 by each of the counting numbers starting with 1. If the quotient is a whole number, the divisor and quotient are a pair of factors.

Dividend	Divisor	Quotient	Factors
72	1	72	1, 72
72	2	36	2, 36
72	3	24	3, 24
72	4	18	4, 18
72	5	14.4	–
72	6	12	6, 12
72	7	~10.29	–
72	8	9	8, 9

The next line would have a divisor of 9 and a quotient of 8. The quotient would be smaller than the divisor, so we stop. If we continued, we would end up only listing the same factors again in reverse order. Listing all the factors from smallest to greatest, we have

$1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36,$ and 72

TRY IT 2.91 Find all the factors of the given number:

96

TRY IT 2.92 Find all the factors of the given number:

80

Identify Prime and Composite Numbers

Some numbers, like 72, have many factors. Other numbers, such as 7, have only two factors: 1 and the number. A number with only two factors is called a **prime number**. A number with more than two factors is called a **composite number**. The number 1 is neither prime nor composite. It has only one factor, itself.

Prime Numbers and Composite Numbers

A prime number is a counting number greater than 1 whose only factors are 1 and itself.

A composite number is a counting number that is not prime.

Figure 2.10 lists the counting numbers from 2 through 20 along with their factors. The highlighted numbers are prime, since each has only two factors.

Number	Factors	Prime or Composite?		Number	Factor	Prime or Composite?
2	1,2	Prime		12	1,2,3,4,6,12	Composite
3	1,3	Prime		13	1,13	Prime
4	1,2,4	Composite		14	1,2,7,14	Composite
5	1,5	Prime		15	1,3,5,15	Composite
6	1,2,3,6	Composite		16	1,2,4,8,16	Composite
7	1,7	Prime		17	1,17	Prime
8	1,2,4,8	Composite		18	1,2,3,6,9,18	Composite
9	1,3,9	Composite		19	1,19	Prime
10	1,2,5,10	Composite		20	1,2,4,5,10,20	Composite
11	1,11	Prime				

Figure 2.10 Factors of the counting numbers from 2 through 20, with prime numbers highlighted

The prime numbers less than 20 are $2, 3, 5, 7, 11, 13, 17,$ and 19. There are many larger prime numbers too. In order to determine whether a number is prime or composite, we need to see if the number has any factors other than 1 and itself. To do this, we can test each of the smaller prime numbers in order to see if it is a factor of the number. If none of the prime numbers are factors, then that number is also prime.

HOW TO

Determine if a number is prime.

Step 1. Test each of the primes, in order, to see if it is a factor of the number.
Step 2. Start with 2 and stop when the quotient is smaller than the divisor or when a prime factor is found.
Step 3. If the number has a prime factor, then it is a composite number. If it has no prime factors, then the number is prime.

EXAMPLE 2.47

Identify each number as prime or composite:

ⓐ 83 ⓑ 77

Solution

ⓐ Test each prime, in order, to see if it is a factor of 83, starting with 2, as shown. We will stop when the quotient is smaller than the divisor.

Prime	Test	Factor of 83?
2	Last digit of 83 is not 0, 2, 4, 6, or 8.	No.
3	$8 + 3 = 11$, and 11 is not divisible by 3.	No.
5	The last digit of 83 is not 5 or 0.	No.
7	$83 \div 7 = 11.857\ldots$.	No.
11	$83 \div 11 = 7.545\ldots$	No.

We can stop when we get to 11 because the quotient $(7.545\ldots)$ is less than the divisor.

We did not find any prime numbers that are factors of 83, so we know 83 is prime.

ⓑ Test each prime, in order, to see if it is a factor of 77.

Prime	Test	Factor of 77?
2	Last digit is not 0, 2, 4, 6, or 8.	No.
3	$7 + 7 = 14$, and 14 is not divisible by 3.	No.
5	the last digit is not 5 or 0.	No.
7	$77 \div 7 = 11$	Yes.

Since 77 is divisible by 7, we know it is not a prime number. It is composite.

TRY IT 2.93 Identify the number as prime or composite:

91

TRY IT 2.94 Identify the number as prime or composite:

137

LINKS TO LITERACY

The Links to Literacy activities *One Hundred Hungry Ants*, *Spunky Monkeys on Parade* and *A Remainder of One* will provide you with another view of the topics covered in this section.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Divisibility Rules (https://openstaxcollege.org/l/24Divisrules)
Factors (https://openstaxcollege.org/l/24Factors)
Ex 1: Determine Factors of a Number (https://openstaxcollege.org/l/24Factors1)
Ex 2: Determine Factors of a Number (https://openstaxcollege.org/l/24Factors2)
Ex 3: Determine Factors of a Number (https://openstaxcollege.org/l/24Factors3)

SECTION 2.4 EXERCISES

Practice Makes Perfect

Identify Multiples of Numbers

In the following exercises, list all the multiples less than 50 for the given number.

215. 2
216. 3
217. 4
218. 5
219. 6
220. 7
221. 8
222. 9
223. 10
224. 12

Use Common Divisibility Tests

In the following exercises, use the divisibility tests to determine whether each number is divisible by 2, 3, 4, 5, 6, and 10.

225. 84
226. 96
227. 75
228. 78
229. 168
230. 264
231. 900
232. 800
233. 896
234. 942
235. 375
236. 750
237. 350
238. 550
239. 1430
240. 1080
241. 22,335
242. 39,075

Find All the Factors of a Number

In the following exercises, find all the factors of the given number.

243. 36
244. 42
245. 60
246. 48
247. 144
248. 200
249. 588
250. 576

Identify Prime and Composite Numbers

In the following exercises, determine if the given number is prime or composite.

251. 43

252. 67

253. 39

254. 53

255. 71

256. 119

257. 481

258. 221

259. 209

260. 359

261. 667

262. 1771

Everyday Math

263. **Banking** Frank's grandmother gave him $100 at his high school graduation. Instead of spending it, Frank opened a bank account. Every week, he added $15 to the account. The table shows how much money Frank had put in the account by the end of each week. Complete the table by filling in the blanks.

Weeks after graduation	Total number of dollars Frank put in the account	Simplified Total
0	100	100
1	$100 + 15$	115
2	$100 + 15 \cdot 2$	130
3	$100 + 15 \cdot 3$	
4	$100 + 15 \cdot [\,]$	
5	$100 + [\,]$	
6		
20		
x		

264. **Banking** In March, Gina opened a Christmas club savings account at her bank. She deposited $75 to open the account. Every week, she added $20 to the account. The table shows how much money Gina had put in the account by the end of each week. Complete the table by filling in the blanks.

Weeks after opening the account	Total number of dollars Gina put in the account	Simplified Total
0	75	75
1	$75 + 20$	95
2	$75 + 20 \cdot 2$	115
3	$75 + 20 \cdot 3$	
4	$75 + 20 \cdot [\,]$	
5	$75 + [\,]$	
6		
20		
x		

Writing Exercises

265. If a number is divisible by 2 and by 3, why is it also divisible by 6?

266. What is the difference between prime numbers and composite numbers?